

SANTOSH PANT

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RESEARCH INTERESTS

Data-efficient geospatial machine learning for climate and agricultural risk in places where ground truth is scarce. I work on evaluation and audit methodology for satellite and drone remote sensing, calibrated uncertainty for decision support, and transfer across regions. Nepal's Terai is my main test bed.

EDUCATION

Knox College

September 2023 to June 2027 (expected)

Bachelor of Science, Computer Science and Data Science (double major); GPA 3.82; Dean's List

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- **Relevant coursework:** Machine Learning; Artificial Intelligence; Algorithm Design and Analysis; Statistics and Probability; Linear Algebra; Differential Equations; Research Seminar and Project

PUBLICATIONS

Accepted

- [1] Santosh Pant. *Auditing GeoFM Evaluation for Field-Extent Segmentation: Label Proxies, Baselines, and When Frozen Features Match Fine-tuning*. Transactions on Machine Learning Research (TMLR), 2026. Single author. Accepted August 2026. openreview.net/forum?id=qRXVTe1yYp; [code: github.com/thesantoshpant/auditing-geofm-evaluation](https://github.com/thesantoshpant/auditing-geofm-evaluation); [artifact: doi:10.5281/zenodo.21993508](https://zenodo.org/record/21993508).
- [2] Santosh Pant. *Calibrated District-Level Prediction Intervals for 2025 Paddy Yield Departure in Nepal's Madhesh Province*. Proceedings of the 34th ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems (SIGSPATIAL 2026), Applications track (4-page paper), Riverside, California, November 2026. Single author. [doi:10.1145/3841645.3843341](https://zenodo.org/record/3841645.3843341); [code and provenance: doi:10.5281/zenodo.22184073](https://zenodo.org/record/22184073).

Under review

- [3] Santosh Pant, Adithya Vaithinathan Asokan, Haoyun Feng, Matthew R. Dennis, Keenan Gibbons, Verrah Akinyi Otiende, and Geoffrey Siwo. *Auditing Split Proximity and Radiometric Scale in RGB-to-Thermal Evaluation*. Submitted to the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Evaluations and Datasets track, August 2026. Under review.
- [4] Santosh Pant, Chau Tran, Dipsha Budhathoki, and Arito Nakamichi. *Crop-Label Alignment: Auditing Crop-Dependent Auto-Labels in Remote-Sensing Segmentation*. Submitted to the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV 2027), Evaluations and Datasets track, August 2026. Under review.
- [5] Santosh Pant. *When Should a Local Model Correct a Global Cropland Map?*. Submitted to the Tackling Climate Change with Machine Learning workshop at NeurIPS 2026, August 2026. Single author. Under review.

RESEARCH EXPERIENCE

Research Collaborator, RGB-to-Thermal Urban Heat Mapping

March to August 2026

University of Michigan Center for Global Health Equity; Mentor: Prof. Geoffrey H. Siwo

Remote

- Continued the second-place UMich heat-resilience hackathon project with the organizing professor. Diagnosed two silent data faults (colormapped JPEG values used as thermal targets, and RGB and thermal frames misaligned by field-of-view mismatch) and rebuilt the pipeline.
- Developed two training-free benchmark checks: a Proximity Audit exposing train-test acquisition-group overlap (on CART, 97.8 to 100 percent of mixing-protocol test frames share a flight with training frames), and an Affine Probe showing that per-frame normalization erases the physical temperature scale before scoring. Led to [3].

Crop-Label Alignment Audit for Remote-Sensing Segmentation

March to August 2026

Knox College; Advisor: Prof. Jurdana Masuma Iqrah

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- Grew out of a Research Seminar project fusing Sentinel-2 imagery with ICESat-2 altimetry for Antarctic sea-ice classification (model implementation, with Chau Tran, Dipsha Budhathoki, and Arito Nakamichi).
- Introduced crop-label alignment, a diagnostic for pipelines that fit label rules inside each image crop; it is identically zero for any crop-invariant predictor. Calibrated it on Sen1Floods11 and applied it to a 17-acquisition sea-ice benchmark, where the model trained on per-patch labels shows a positive alignment contrast against a scene-label control in all 17 folds. Led to [4].

Sparse-Label Cropland Mapping, Nepal Terai

2026

Independent research, Knox College

Galesburg, Illinois

- Replicated the Kerner et al. (NeurIPS 2020 CCAI workshop) multi-headed LSTM cropland pipeline on Sentinel-2 time series, extended it to Nepal's Terai with sparse CropHarvest labels, and documented a label-source confound in the public data.

- Reframed the question as when a local model should be allowed to correct a global cropland map (ESA WorldCereal). Derived metric-specific break-even conditions for map edits, showed that conclusions flip between F0.5 and F2, and found that blind edits move F1 in both directions across seven datasets. Proposed a stratified label screen before editing. Led to [5].

Conformal Prediction Intervals for District Paddy-Yield Loss, Madhesh Province

2026

Independent research, Knox College

Galesburg, Illinois

- Built leak-controlled district-level prediction intervals for Nepal's 2025 Madhesh drought from a five-model LightGBM ensemble with split and covariate-shift-weighted conformal calibration on Earth Engine features, plus a Bihar transfer diagnostic; the 2025 output was 0 screen-positive and 8 inconclusive across eight districts. Led to [2].

Geospatial Foundation-Model Evaluation Audit

2025 to 2026

Independent research, Knox College

Galesburg, Illinois

- Audited how label proxies and baselines change geospatial foundation-model rankings for field-extent segmentation on single-date Sentinel-2 imagery from six countries (Fields of The World). Replacing field polygons with ESA WorldCover cropland raises random-forest AUROC by up to 0.34, and a from-scratch U-Net beats a Prithvi frozen probe in five of six regions.
- Compared frozen-backbone decoders with full fine-tuning for Prithvi-EO-2.0 and TerraMind over ten matched seeds: TOST-equivalent within 0.02 AUROC in 9 of 12 region-by-model cells, Kenya (about 0.1 percent positive pixels) the common failure, and the frozen Prithvi decoder trains about 110 times fewer parameters. Led to [1].

Satellite Heat-Stress Classification, Nepal Terai

2026

Independent research, Knox College

Galesburg, Illinois

- Built an early-season vegetation heat-stress classification pipeline over 21 Terai rice districts in Google Earth Engine (MODIS LST and NDVI, ERA5-Land, ESA WorldCover; 59,793 pixel-years) and deployed a live forecast app (terai-heat-forecaster.streamlit.app).
- Under year-blocked leave-one-district-out evaluation with a paired district-block bootstrap, previous-year persistence outperforms a fold-safe XGBoost ensemble on average (macro-F1 0.476 vs 0.362); proposed persistence and modal-class nulls as required baselines for sub-national benchmarks.

SELECTED SOFTWARE PROJECTS

CropScan: calibrated crop-disease diagnosis for small growers

2026

PyTorch (DINOv2 with LoRA, EfficientNetV2-S), FastAPI, React, MongoDB, Docker; three-person team: I built the ML pipeline and most of the backend

- A leaf-validity gate followed by a calibrated two-model ensemble that commits only when both models agree and otherwise asks for a better photo. Trained on 69,674 images across 38 diseases and 14 crops; 75.6 percent accuracy on real field photos (PlantDoc) against 99.7 percent on lab images, reported openly. Shaped by 30+ user interviews at Knox Farm. Live at cropscan.tech.

HONORS AND AWARDS

- **Richter Research Scholar, Knox College:** four funded research terms, 2024 to 2026.
- **Break Through Tech AI Fellow, Cornell Tech:** 2026 to present.
- **HackIllinois 2026 (University of Illinois Urbana-Champaign):** 1st place, Best Use of Actian VectorAI DB (sponsor track).
- **University of Michigan AI for Urban Heat Resilience Hackathon 2026:** 2nd place (\$2,000 prize), for Game of Drones.
- **ASA DataFest 2026 (Central Illinois):** Best Insights Award, team Outliers, survival analysis of 200,000 health records.
- **ICPC North Central Regional,** participant (UIUC site), 2025.

TEACHING AND SERVICE

- Teaching Assistant, Data Structures and Algorithms (two terms); Computer Science Tutor and Computer Science Club lead, Knox College.

TECHNICAL SKILLS

- **Geospatial and remote sensing:** Google Earth Engine, Rasterio, geopandas, Sentinel-1/2, MODIS, Landsat, ERA5, CHIRPS, ESA WorldCover, ESA WorldCereal, Fields of The World, CropHarvest, drone RGB and thermal imagery
- **Machine learning:** PyTorch, scikit-learn, XGBoost, LightGBM, EO foundation models (Prithvi-EO-2.0, TerraMind, Clay, AnySat), CNN and U-Net segmentation, LSTM time-series models
- **Methods:** split and covariate-shift-weighted conformal prediction, TOST equivalence testing, leakage-aware spatial and temporal evaluation, block bootstrap, preregistration and adversarial self-review
- **Languages and tools:** Python, R, Go, TypeScript, SQL; FastAPI, Streamlit, Folium, React, PostgreSQL, Docker, Git, Linux, pytest